

2 A socio-geographical investigation of lexical variability in England

Evidence from the English Dialects App

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2.1 Introduction

The last systematic nationwide investigation of dialect variation in England was the *Survey of English Dialects (SED)* (Orton et al., 1962–1971), dating from the 1950s and 1960s, investigating a demographically extremely restricted subsample of the population – the now famous Non-Mobile Old Rural Men (NORMs – Chambers & Trudgill, 1980). This extraordinary survey presents a picture of the (apparently) most non-standard traditional speakers in England, born in the second half of the 19th century. Since then, theoretical and empirical fashions in the discipline of dialectology have changed, and there has been a much greater focus on individual variables studied in individual sites, in individual communities within those sites or even within the speech of lone individuals. We have little recent sense, therefore, of the contemporary *national* picture of dialect variation as opposed to the local one.

In order to try to address this, and inspired by the Swiss *Dialäkt Äpp* (Kolly & Leemann, 2015), we developed the *English Dialects App (EDA)*, a smartphone application that was able to remotely gather data that could shed light on phonological, morphosyntactic and lexical variation. A relatively early example of the use of smartphones to collect dialect data, it has been joined over the past decade by a range of other internet-reliant strategies to harvest large dialect datasets. These include analyses of data from the social media platform *Twitter* (e.g. Ilbury et al. 2024; Stevenson 2024; Willis 2020), as well as other online or browser-based surveys (e.g. Avanzi 2017; MacKenzie et al. 2022). Such technologies not only make it possible to source data from a broad cross-section of the community across a wide geographical area (Blaxter & Britain 2021) but are also extremely cost-effective in comparison with traditional dialectological fieldwork techniques and allow data to be collected, processed and analysed quickly.

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In addition, thanks to their adoption of widely used social media technologies and the attention they have gathered from the more traditional TV, radio and newspaper outlets, they have been successful in bringing language and dialect issues to a wider audience in an accessible way (Avanzi 2017; Britain et al. 2018). We begin by briefly introducing the *EDA* (see Leemann et al. 2018) before presenting what our analysis of the App tells us about three lexical variables – AUTUMN, SNAIL and SPLINTER – and their social and geographical distribution in England.

2.2 The English Dialects App (*EDA*)

The *EDA* was a free smartphone application with two main functions – a dialect quiz and dialect recordings. The quiz asked people which variant of 26 different phonological, morphosyntactic and lexical variables they themselves use. These 26 variables were deliberately chosen to enable direct comparisons with the *SED* data and to allow us to track how language had changed in the intervening half century. For the dialect recording function, users were able to upload recordings of themselves reading a short version of the *Boy Who Cried Wolf* story and could also listen to already uploaded recordings, those submitted by other app users, as well as recordings of speakers from the *SED*.

In the application, launched in early 2016 on both iOS and Android platforms, each screen in the quiz prompted users to either answer a question or complete a fill-the-gap task. Below the questions, users could read the possible answers and click on whichever one they thought they themselves use. The app then showed a map of where in the country their answer was also found.¹ The app then moved on to ask the next question. Once the quiz had been completed, users were directed to our ethics and privacy policy and were asked to accept the terms and conditions. If they agreed, they were asked to locate themselves geographically using a pin on a map and to submit metadata about their social characteristics, their educational achievements and their mobility levels (Britain et al. 2018, 2020, 2021; Leemann et al. 2018). The combination of national coverage and speaker social metadata enabled us to investigate not only *where* certain variants were being used but also *what kinds of people* were mostly using them within those areas. In other words, it provided, on the one hand, the geographical coverage that much contemporary dialectological work does not, without losing sight, on the other hand, of the social diversity of the population, a consideration ignored in the earlier *SED*. Users' quiz answers and social background information were then anonymously stored on a server. By the end of 2018, over 50,000 people, from more than 4,900 localities across the UK, had answered the quiz and added their social background metadata. In addition, over 4,000 people uploaded recordings of themselves

reading the *Boy Who Cried Wolf*. We focus here, however, on the quiz data, since the reading passage recordings were not designed for the investigation of lexical variability (see Leemann et al. 2018 for more details).

We are, of course, fully aware of the drawbacks of collecting dialect data in this way. The sample gathered with the app is biased towards digital natives. We bear in mind that, in a sense, *EDA* data, when looked at in their entirety, cast perhaps an overly pessimistic view on the survival of traditional dialect features, given the sample skew towards the young, the urban and the educated. The *SED*, however, it must be admitted, therefore shines an overly optimistic light on the fate of the same features, given the extreme skew towards the old, the rural and the less educated – a skew that was deliberate in order to target those who the researchers believed would produce the best preserved ‘traditional types of vernacular English’ (Orton 1962: 14). One advantage of the *EDA* is that, despite the overall skew, we can control for user age, educational level, mobility and so on, given that we collected social metadata. We are also aware that the quiz is not collecting dialect *use* but, ultimately, *intuitions* about dialect use (see Labov 1996). We present an honest account of these drawbacks elsewhere (Leemann et al. 2018). One strength of the now outdated *SED* was the possibility of observing patterns of variation across the whole country, not possible in contemporary social dialect investigations of individual localities. One strength of these locally oriented social dialect studies, however, is that they can probe how variability differs according to the age, gender, educational level and mobility of the users, something the *SED* could not do. The *EDA* can do both – geographical coverage and at least some degree of social depth, enabling a simultaneous social and geographical perspective on variability.

2.3 Analysis

The App quiz contained three lexical variables for which we harvested data: variation in {AUTUMN}, {SNAIL} and {SPLINTER}. These three lexical items had been included in the *SED*, and thus we were able to directly compare evidence collected in the *EDA* with maps derived from the *SED* data. We present the specific analysis and results for each lexical variable in separate sections below. In each case, we first present the state of affairs as logged in the *SED*, to show the distribution of the main variants of these variables as used among NORMs in the mid-20th century. We then compare the *SED* distributions with those gathered from the *EDA*. Note that in maps contrasting the *SED* with the *EDA*, the *SED* maps plot the (usually) one token gathered from each of the 300+ locations nationwide in the *SED*, whereas the *EDA* plots are based on 45,287 tokens from the 4,000+ localities in England. Different mapping techniques are therefore

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used for mapping the *SED* data (essentially dot plots, where each location is represented by one circle on the map, usually showing how the one respondent from that location answered the question) than the *EDA* data. Given the amount of data in the *EDA*, we are able to present gradient frequency maps, rather than simply categorical measurements for each location.²

For the sociolinguistic analysis, we then identified those areas of the country where concentrations of specific variants were found in the *EDA* data and then zoomed in for a closer investigation of social variability. In each of these focus areas, we used mixed-effects regression analysis to model whether variant choice significantly interacted with a range of social factors, information about which we collected from each app user: age, gender; ethnicity; mobility (how often has the user moved in the last ten years; how far do they commute to work or school); and highest educational qualification. This enabled us to establish the social profiles of those who continued to use the traditional variants under scrutiny.

2.4 Results

2.4.1 AUTUMN

The investigation of AUTUMN was motivated partly by an interest in the extent to which the most widespread traditional dialect forms – *fall* and *backend* – were still being used, where and by whom. We were also intrigued to explore journalistic claims of a reported Americanisation of British English and the extent to which *fall*, originally a British form exported to the USA, might be making a comeback via American influence. Is *fall*, therefore, only used in its traditional dialect heartlands or also being used elsewhere and, if so, where and by whom? The *SED* identified a number of different variants of AUTUMN as a result of presenting the following prompt to its informants: ‘If (the event happened) in the season after summer? (You’d say: it happened) in ...’³ (Orton & Wakelin 1967 828). Answers included *autumn*, *backend*, *fall*, *latter end*, *harvest time* and *after season*. Since *latter end*, *harvest time* and *after season* were used by very few informants, we focus our attention here on the three main variants *autumn*, *backend* and *fall*. The *SED* shows that *backend* is essentially a northern form: it is found right across the North of England, except Lincolnshire, and stretches in the west down along the border with Wales, and also into the North and East Midlands, into Leicestershire, Nottinghamshire, Derbyshire, Staffordshire, Cheshire and Shropshire. It is almost entirely absent from the South. Meanwhile, *fall* is concentrated in Lincolnshire, on the one hand, but more prominently in the South-West of England, especially in the areas of Somerset, Gloucestershire and Dorset within reach of Bristol, on the other hand. There are also small concentrations in south Devon and

Cornwall in the South-West and in Bedfordshire and Buckinghamshire to the north-west of London. Note that the London area, the urban areas of the Midlands, around Birmingham, and the North-West, around Manchester and Liverpool, show no use at all of *fall*. The rest of England, with very few exceptions, used *autumn* (see Figures 2.1 and 2.2).

In the *EDA*, we asked: ‘What is the season that follows summer?’ providing the three main options found in the *SED*, *backend*, *fall* and *autumn*.

The results for *backend* in the *EDA*, alongside those from the *SED* to facilitate the contrast, can be seen in Figure 2.1. Comparing the two, we observe that:

- a Nowhere is *backend* used as a majority community variant – the highest concentrations, at around 12%, were found in and around Bridlington in East Yorkshire.⁴
- b The area in which *backend* is found has shrunk, relative to the *SED*. The *backend* zone has retreated northwards: it is now largely absent from the Midlands and is only used at extremely low levels in the lower North, except, surprisingly, in central Lincolnshire, where, by contrast, *fall* had been common in the *SED*.

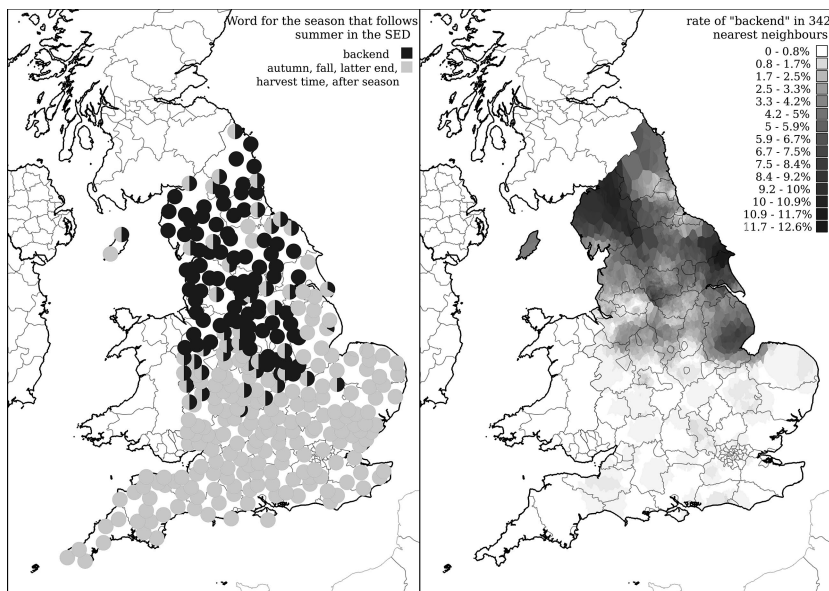


Figure 2.1 The geographical distribution of *backend* in the *Survey of English Dialects* (Orton et al. 1962–1971), left, and the *English Dialects App* (N = 45,287), right.

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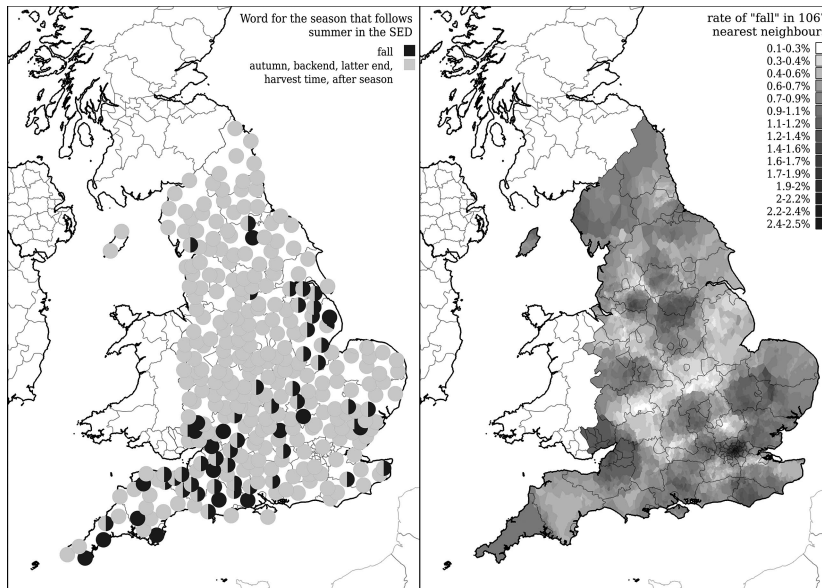


Figure 2.2 The geographical distribution of *fall* in the *Survey of English Dialects* (Orton et al. 1962–1971), left, and the *English Dialects App* ($N = 45,287$), right.

- c It is most common in the eastern coastal areas of the Yorkshire counties, around Huddersfield in West Yorkshire, and in Cumbria, in the North-West, especially around Carlisle.

For the sociolinguistic analysis, we then zoomed in on these Northern counties where levels of *backend* were higher than elsewhere and examined the social profile of its users. Table 2.1 shows the significant factors. Perhaps not surprisingly, age is highly significant – the older the app user, the more likely they are to offer *backend* as their lexical choice for AUTUMN. Non-mobility also promotes the traditional dialect form – those users who had not moved house in the past ten years were more likely to claim use of *backend*. Non-binary gender was also flagged as significant, based however on a very low number of individuals.⁵ Men were also more likely to use *backend* but not significantly so. No other factors showed significance.

The results for *fall* in the *EDA*, alongside those from the *SED* to facilitate the contrast, can be seen in Figure 2.2. Our comparison shows that:

- a *Fall* has almost entirely disappeared from the dialect map of England. No community has levels of *fall* use above 3%.

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Table 2.1 Best fit model for social factors and the use of *backend* in Northern England from the *English Dialects App* (N = 7,353)

	Coefficient	Std. error	Pr (> t)	Sig.
(Intercept)	−0.0270962	0.0074502	0.000278	***
Age	0.0017458	0.0001718	<2e-16	***
Gender: male	0.0048186	0.0047914	0.314605	
Gender: other	0.1734633	0.0643676	0.007057	**
House moves: four times or more	0.0025517	0.0067701	0.706253	
House moves: I still live in the same home as ten years ago	0.0215562	0.0065764	0.001051	**
House moves: once	0.0073748	0.0065870	0.262918	

Significance codes: *** < 0.001, ** < 0.01, * < 0.05. Residual standard error: 0.2032 on 7,292 degrees of freedom (DF). Multiple R-squared: 0.01901. Adjusted R-squared: 0.0182. F-statistic: 23.55 on 6 and 7,292 DF. *p*-value: < 2.2e-16. Reference levels: female, two to three house moves in the past ten years.

- b Perhaps surprisingly, the highest levels of use in the country are not found in its traditional dialect heartlands of the South-West but in central London, in the Boroughs of Tower Hamlets, Hackney, Southwark and City of London, where levels of 2%–3% are found.
- c The traditional focal areas of the South-West are still home to visibly more frequent users of *fall* than most others, especially around Bristol.
- d Other ‘hotspots’ (or perhaps we should call them warm spots, since levels are still very low) include Sheffield, Manchester, East Sussex, and the parts of West Suffolk and South-West Norfolk in the vicinity of very large American Airforce bases at Mildenhall and Lakenheath.

The *EDA* data show ~~that~~ not only a shrinkage of the original *fall* area in the South-West but also its (very subtle) appearance in areas where it had not been noted in the *SED*. In order to explore the social profiling of *fall* use, we examine it in two very different locations – one in its original heartland, Bristol, and one in an urban area where it was entirely absent in the *SED* data, namely central London. Figure 2.3 shows the areas under closer scrutiny for *fall* and the distribution of this variant within them.

The *EDA* data from Bristol (in Table 2.2) were sufficient to examine age, gender and ethnicity, but only age showed an effect, with younger speakers slightly more likely to use *fall* than older. In the central London data (in Table 2.3), we found that those users who identified as Asian, or of an ethnicity other than White, Black, Asian or Mixed, were more likely to use *fall* than others, while the other social factors were not significant.

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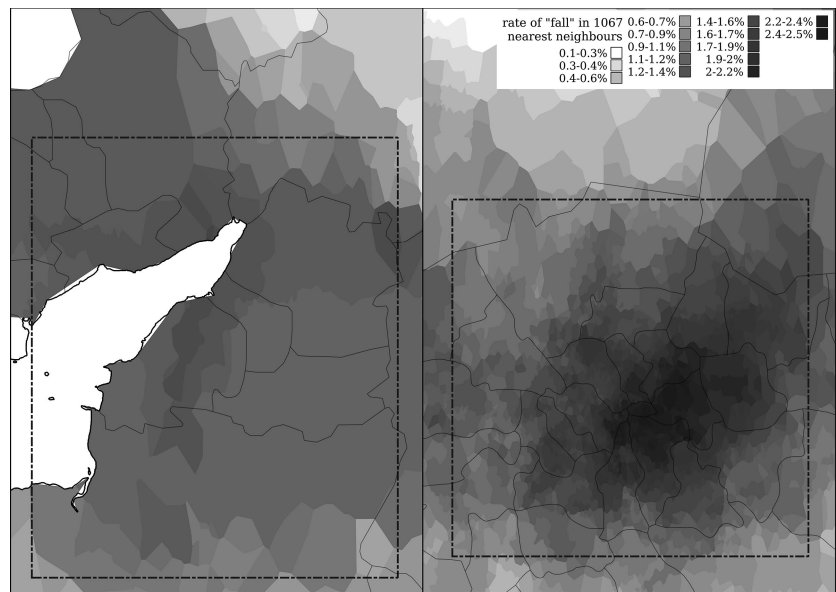


Figure 2.3 The geographical distribution of *fall* in Bristol ($N = 1,249$), left, and central London ($N = 3,894$), right, from the *English Dialects App*.

Table 2.2 Best fit model for social factors and the use of *fall* in Bristol, South-West England, from the *English Dialects App* ($N = 1,249$)

	Coefficient	Std. error	Pr(> t)	Sig.
(Intercept)	0.0171805	0.0297300	0.56345	
Age	-0.0006087	0.0002126	0.00426	**
Gender: male	-0.0048612	0.0059999	0.41797	
Gender: other	-0.0200874	0.0745095	0.78752	
Ethnicity: Black	0.0053371	0.0673769	0.93688	
Ethnicity: mixed	0.0555850	0.0382793	0.14673	
Ethnicity: other	0.0036384	0.0554861	0.94773	
Ethnicity: White	0.0162991	0.0293911	0.57930	

Significance codes: *** < 0.001, ** < 0.01, * < 0.05. Residual standard error: 0.1052 on 1,239 degrees of freedom (DF). Multiple R-squared: 0.01026. Adjusted R-squared: 0.004671. F-statistic: 1.835 on 7 and 1,239 DF. p -value: 0.07694. Reference levels: female, Asian.

2.4.2 SNAIL

Our consideration of SNAIL was driven by the desire to examine lexical variants we knew, from earlier work, to be undergoing obsolescence and to use the affordances of the *EDA* to examine this obsolescence over apparent-time

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Table 2.3 Best fit model for social factors and the use of *fall* in central London from the *English Dialects App* (N = 3,894)

	<i>Coefficient</i>	<i>Std. error</i>	<i>Pr(> t)</i>	<i>Sig.</i>
(Intercept)	0.0159679	0.0055833	0.00426	**
Age	–0.0001016	0.0001284	0.42881	
Gender: male	–0.0037027	0.0037846	0.32795	
Gender: other	–0.0238958	0.0476739	0.61623	
Ethnicity: Asian	0.0285686	0.0077493	0.00023	***
Ethnicity: Black	–0.0034757	0.0112485	0.75734	
Ethnicity: mixed	0.0024202	0.0083276	0.77135	
Ethnicity: other	0.0695718	0.0119478	6.25e-09	***

Significance codes: *** < 0.001, ** < 0.01, * < 0.05. Residual standard error: 0.1164 on 3,880 degrees of freedom (DF). Multiple R-squared: 0.01294. Adjusted R-squared: 0.01116. F-statistic: 7.266 on 7 and 3,880 DF. *p*-value: 1.118e-08. Reference levels: female, White.

by comparing *EDA* users of different ages. Lexemes denoting *SNAIL* that contained *–dod–*, such as *hodmedod*, *hoddydod*, *dodman*, *dodderman*, were used at least sporadically in many parts of southern and central England back in the early 19th century (Wright 1906), but by the time of the *SED* they were largely restricted to Eastern England (see Britain 2009).

The *SED* identified a number of different variants of *SNAIL* as a result of asking the following question of its informants: ‘What are those slow slimy things that carry their houses about with them; they come out after rain?’ (Orton 1962: 67). Answers included *snail*, *dodman* (and other variants with *dod–*), *hodmedod* (and other variants with *hod–*), *packman*, *snag*, *sniggle*, *snarlbug* and *bullhorn*. Since the latter five were, in total, used only 12 times in the entire *SED* and were often explicitly listed as ‘older’ forms (e.g. *snarlbug* in Brompton Regis in Somerset (Orton & Wakelin 1967: 478)), we focus our attention here on the three main variants *snail*, *dodman* and *hodmedod*. Figure 2.4 shows the distribution of *SNAIL* in the *SED* data. Variants of *dod–* are entirely concentrated in Norfolk. Variants of *hod–* are concentrated in Suffolk and Essex, to the immediate south of Norfolk. The main non-*snail* variants are, therefore, essentially East Anglianisms.

In the *EDA*, we asked: ‘Animals that carry their house on their back are___?’, with three possible answers: *hodmedods/hoddy-dods/hoddy-doddies*; *dodmen*; *snails*. The results from the *EDA* can be seen in Figure 2.5. We observe that:

- a Levels of use of both *dodman* and the various *hod–* forms have fared badly since the *SED* – the highest levels in the region are of *dodman* in East Norfolk at just 5%.

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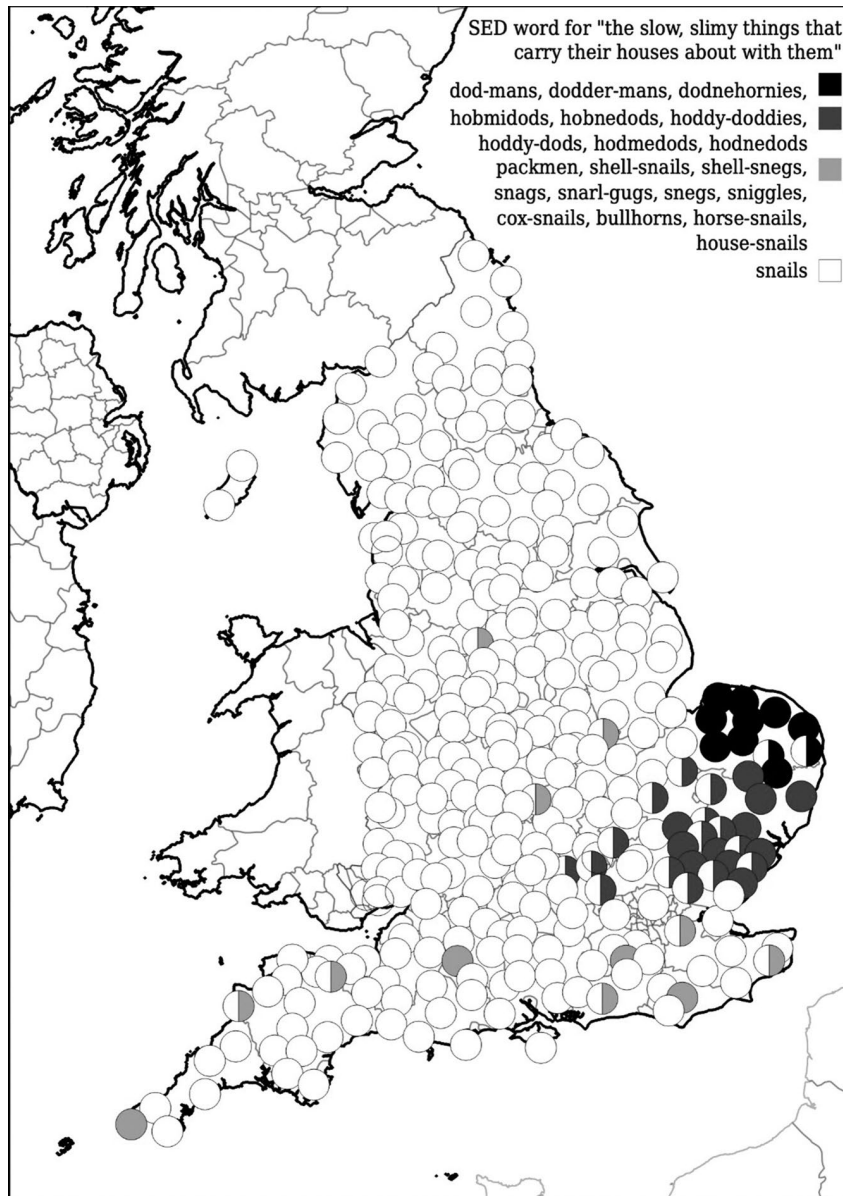


Figure 2.4 The geographical distribution of variants of SNAIL derived from the *Basic Materials of the Survey of English Dialects* (Orton et al. 1962–1971).

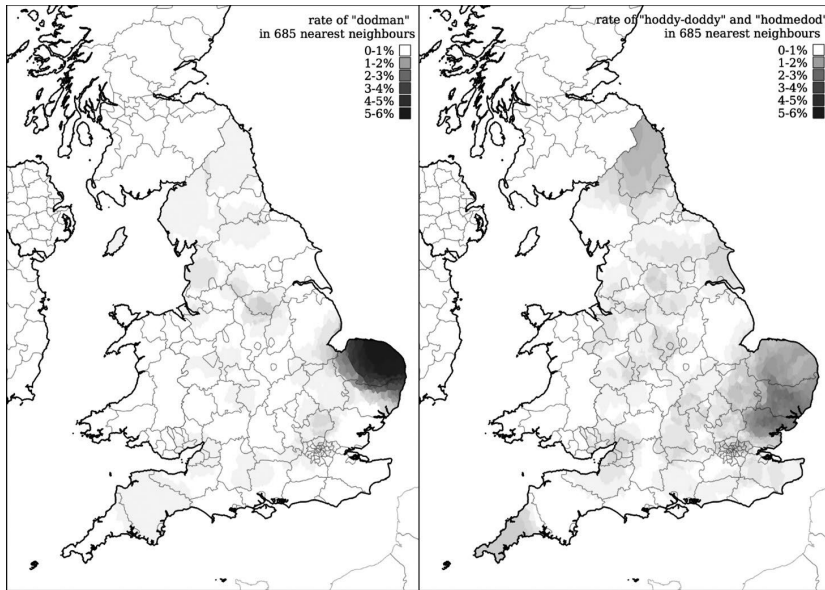


Figure 2.5 The geographical distribution of *dodman*, left, and *hodmedod/hoddy-dod(dy)*, right, in the *English Dialects App* (N = 45,287).

- b *Dodman* has fared better than the *hod-* forms to the south. The *hod-* area has shrunk in geographical spread since the *SED* and is now largely restricted to Suffolk and North Essex, though it is notably also a minority form to even *dodman* in Norfolk. This is perhaps not surprising given that the *hod-* area is nearer to the dialect levelling influences of London than the *dodman* area.
- c Results from the far north-east of Suffolk, around Lowestoft, showing a preference for *dodman* over *hod-* forms, lend added weight to Trudgill's (2021: 57) claim that this corner of Suffolk is part of linguistic Norfolk.

Given the higher levels of the traditional form in Norfolk, we zoom in on the use of *dodman* there for closer sociolinguistic inspection. Table 2.4 shows that age and level of education are significant predictors of the use of *dodman*. Gender, ethnicity and mobility have no significant effect.

Within this dataset, there is a linear relationship between speaker age and *dodman* usage: older speakers are significantly more likely to report using *dodman* (see Figures 2.6 and 2.7). While in some areas of Eastern Norfolk, levels of *dodman* use reach almost 7% among the older users,

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Table 2.4 Best fit model for social factors and the use of *dodman* in Norfolk from the *English Dialects App* (N = 585)

	Coefficient	Std. error	Pr(> t)	Sig.
(Intercept)	0.0731300	0.0588096	0.21419	
Age	0.0041013	0.0007348	3.69e-08	***
Highest qualification: A levels, Bac, vocational level 3, etc.	-0.1446061	0.0556676	0.00963	**
Highest qualification: higher education (BA, BSc, MA, PGCE, etc.) and professional/ vocational equivalents	-0.1555302	0.0514169	0.00260	**
Highest qualification: 5 GCSE grade A*-C, 5 O-Levels, vocational level 2, etc.	-0.1429064	0.0598358	0.01725	*
Highest qualification: I am not yet 16 or haven't yet taken GCSEs	-0.1325993	0.0764344	0.08332	
Highest qualification: no qualifications	-0.2895893	0.0951315	0.00244	**
Highest qualification: other (e.g. foreign qualifications)	-0.2402593	0.1310888	0.06735	

Significance codes: *** < 0.001, ** < 0.01, * < 0.05. Residual standard error: 0.2427 on 569 degrees of freedom (DF). Multiple R-squared: 0.07826. Adjusted R-squared: 0.06692. F-statistic: 6.901 on 7 and 569 DF. *p*-value: 6.768e-08. Reference level: fewer than 5 GCSE grade A*-C/fewer than 5 O-Levels.

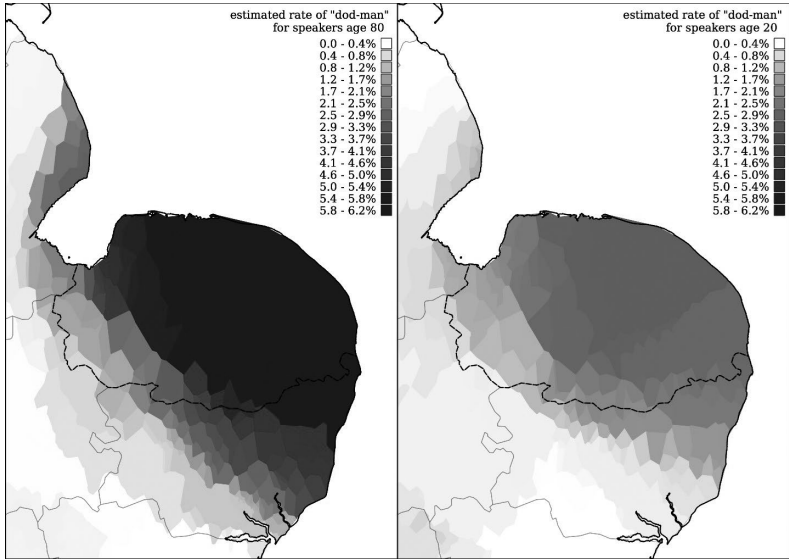


Figure 2.6 The geographical distribution in the *English Dialects App* of *dodman* in older, left, and younger, right, users.

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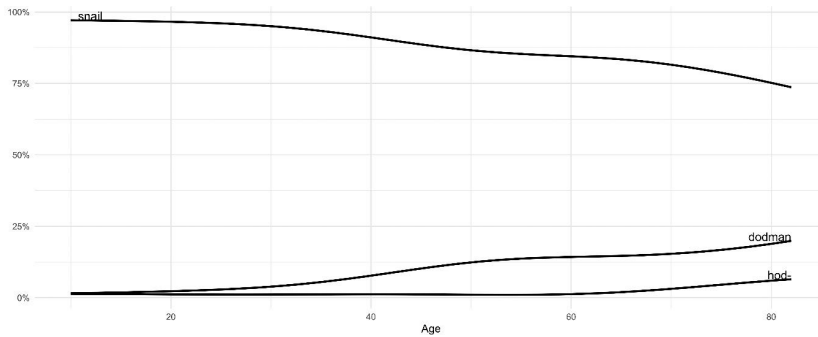


Figure 2.7 The use of *snail*, *dodman* and *hod-* variants of SNAIL by age in the *English Dialects App* data from Norfolk.

it nowhere exceeds 3.5% among the young. For highest level of education, users with ‘Fewer than 5 GCSE grade A*-C, or fewer than 5 O-Levels’ are significantly more likely to use *dodman* than those with higher levels of education.

2.4.3 SPLINTER

In the *SED*, in response to the question ‘What do you call a very small piece of wood that has got into your finger?’ (Orton 1962: 78), over 20 different lexemes were found, the main ones of which can be seen in Figure 2.8. As with SNAIL and AUTUMN, we are interested here to see whether such variety survives in the dialects of England today and to explore the socio-demographic profiles of those who retain traditional variants. The *EDA*, in response to the question ‘A small piece of wood stuck under the skin is a ____’, with possible responses *spool*, *spile*, *speel*, *spell*, *spelk*, *shiver*, *spill*, *sliver*, *splinter* and *splint*, found that, today, *splinter* was the majority form everywhere except in the North-East of England, where *spelk* remains the dominant form (see Blaxter & Britain 2021). Many of the traditional variants remain, however, as minority forms in certain parts of the country – one of these is *spell*, which was used at the time of the *SED* across Yorkshire and parts of Lancashire and Cumbria in Northern England. We focus here, then, on where *spell* is still used, to what degree and the socio-demographic profiles of typical *spell* users. Figure 2.9 compares the data derived from the *SED* with the *EDA* data from around Yorkshire, the *spell* heartland. The area of *spell* use in the *EDA* corresponds almost entirely with that of the NORMs of the *SED* collected 60 years earlier – there appears to be little geographical shrinkage in its spread. What we do see

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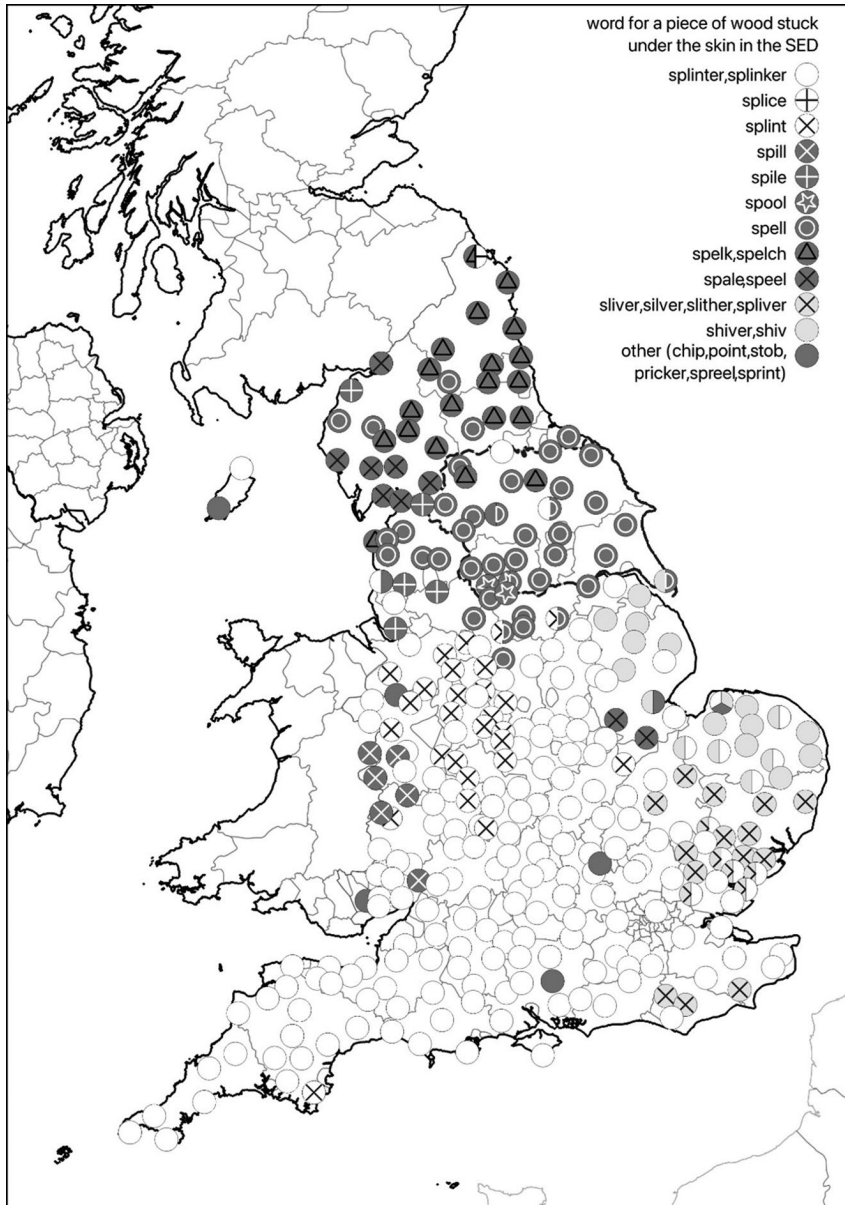


Figure 2.8 The geographical distribution of variants of SPLINTER in the *Survey of English Dialects* (Orton et al. 1962–1971).

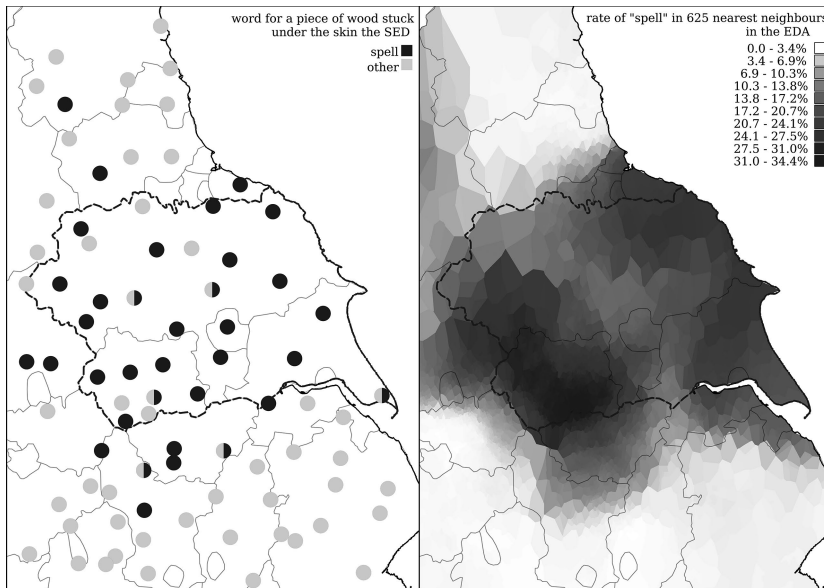


Figure 2.9 The distribution of *spell* tokens of SPLINTER in Yorkshire and Humberside in the *Survey of English Dialects*, to the left, and *English Dialects App*, to the right.

in Figure 2.9 is that the proportion of the population that uses *spell* is relatively low – the communities in West Yorkshire with the highest concentrations of *spell* only reach levels around 35%, and it is found even less frequently elsewhere.

In order to reveal the social profile of *spell* users within this core area, we looked only at EDA data in the East Riding of Yorkshire, North Yorkshire, West Yorkshire and Hull, excluding the few users who report forms other than *splinter* or *spell*. The results are shown in Table 2.5. As with SNAIL in Norfolk, level of education and age are significant predictors of use: gender, ethnicity and mobility have no significant effect.

Within this dataset, there is a linear relationship between speaker age and *spell* usage: older speakers are more likely to report using *spell*. Figure 2.10 shows quite dramatically how *splinter* has taken over from *spell* in this area. The majority form is *splinter* for those under 57 years old, and among the youngest speakers *spell* use is below 10%. Speakers without university education use *spell* significantly more often than those with, though the effect is not an especially strong one.

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Table 2.5 Best fit model for social factors and the use of *spell* in Yorkshire from the *English Dialects App* (N = 4,386)

	Coefficient	Std. error	Pr(> t)	Sig.
(Intercept)	-0.235196	0.021340	<2e-16	***
Age	0.013792	0.000532	<2e-16	***
Highest qualification: fewer than 5 GCSE grade A*-C or fewer than 5 O-Levels	0.078502	0.038067	0.0393	*
Highest qualification: 5 GCSE grade A*-C, 5 O-Levels, vocational level 2, etc.	0.056523	0.028455	0.0471	*
Highest qualification: A levels, Bac, vocational level 3, etc.	0.037594	0.019536	0.0544	
Highest qualification: I am not yet 16 or haven't yet taken GCSEs	0.094111	0.038602	0.0148	*
Highest qualification: no qualifications	0.143515	0.072801	0.0488	*
Highest qualification: other (e.g. foreign qualifications)	-0.050189	0.109463	0.6466	

Significance codes: *** < 0.001, ** < 0.01, * < 0.05. Residual standard error: 0.3934 on 3,170 degrees of freedom (DF). Multiple R-squared: 0.195. Adjusted R-squared: 0.1932. F-statistic: 109.7 on 7 and 3,170 DF. *p*-value: <2.2e-16. Reference level: higher education (BA, BSc, MA, PGCE, etc.) and professional/vocational equivalents.

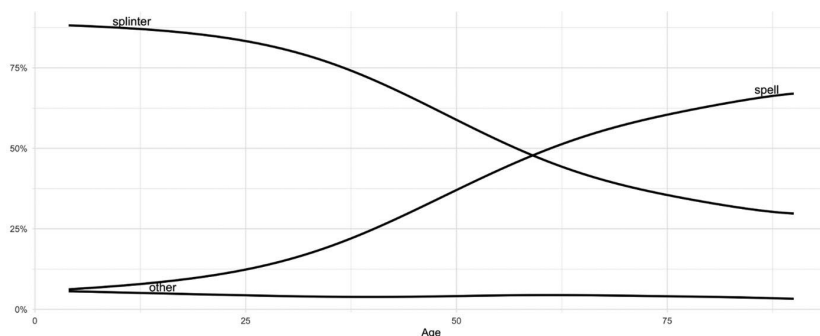


Figure 2.10 Age and the distribution of *spell* and *splinter* tokens of SPLINTER in North, West and East Yorkshire, as well as Hull, from the *English Dialects App*.

2.5 Discussion

Investigating these three lexical variables in the EDA has highlighted the power of using smartphone apps to collect data on lexical variation and change and the dramatic extent of lexical levelling since the *SED* and

flagged some sociolinguistic commonalities within the social profiles of traditional lexeme users. The *SED*, as mentioned earlier, presents us with a rather optimistic view of the health of lexical variation in the mid-20th century, capturing as it does the usage of those most likely to deploy traditional dialect forms. It does not capture social variability, so we do not know what younger or more mobile speakers were doing at that time. We can be pretty certain that at least some of these lexemes were already in the early stages of attrition even then, but the *SED* data do not capture that shift. The *EDA*, relatively cheaply and efficiently, is able to give us a nationwide, socially differentiated snapshot of variability today across different age groups – the apparent-time perspective enables us to track obsolescence in progress, the geographical perspective enables us to observe changes in the spatial distributions of variants and the social perspective enables us to see the demographic profiles of those who retain traditional dialect forms and those who are opting for the supralocal, more widespread forms.

The *EDA* presents an especially pessimistic picture for the lexical items considered, suggesting dramatic dialect levelling. All of the features that were majority forms in their heartlands at the time of the *SED* are now at best minority forms and at worst clinging on to existence. The lexeme *spell* in Yorkshire is faring best: it is still a majority form there among the retired, but among the young its use is now rare. Unlike the other features considered in this analysis, its geographical scope remains largely the same as at the time of the *SED*, just at considerably lower levels. In Norfolk, the use of *dodman* for *SNAIL* has been pushed back to the East of the county, and even among the older speakers it is found at very low levels indeed. The *backend* variant of *AUTUMN* has also undergone geographical shrinkage retreating away from areas of the Midlands where it was commonly found at the time of the *SED* and is now found only in parts of the North of England, again, like *dodman* at much lower levels. Our examination of *fall* suggests that it has more or less disappeared in its original heartlands of the South-West, being found at its most ‘vibrant’ at levels of just 2%–3% in urban Bristol and less in the surrounding rural areas. Interestingly, this urban-rural divide, whereby the traditional form is found at higher levels in the city than in the countryside, was also found for rhoticity in the *EDA*, a reflection, we speculate, of decades of middle class counterurbanising migration from London and South-East into the more rural parts of the South-West of England (Grossenbacher 2016; Leemann et al. 2018). The appearance of *fall* in London, especially among the Asian and other minority ethnic communities there, is at such low levels that it is premature to assume it is making a comeback as a result of Americanisation.

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In our investigation of specific traditional lexemes, we ran mixed effects regression analyses, considering as many social constraints – age, gender, ethnicity, mobility and education – as feasible given the number of app users in each region. Apart from for *fall*, which, as we have seen, appears almost extinct, age was an extremely significant factor affecting the vitality of all the lexemes, reiterating even across apparent-time the dramatic levelling underway. Gender was not a significant factor reliably constraining any of the lexical items, and nor, apart from for obsolescent *fall*, was ethnicity. Education was significant for *spell* and *dodman* as was home-moving frequency for *backend*, suggesting an important role for social mobility and class in driving levelling – the more mobile, the more educated, the more levelled. In the context of increased local and supralocal mobility, alongside wider participation in higher levels of education over the period since the *SED* (Britain 2010, 2013), these four traditional regional dialect lexemes have, not surprisingly, fared badly, a fate that the *EDA* has been able to track socially, geographically and diachronically.

Notes

- 1 At first, these plotted dialect distributions from the *SED*; from May 2016 they displayed maps based on answers from earlier users of the *EDA*.
- 2 All plots used to present the *EDA* data show interpolation and smoothing using kernel smoothing with an adaptive square kernel (i.e. *k*-nearest neighbour smoothing). This is a technique that smooths raw figures at a particular location by averaging the score at that location with those at the others but incorporating a calculation that weights the corresponding scores in nearby locations more than those ever further away (for more on such methods, see Blaxter 2017, 2019; Rumpf et al. 2009; Sibling et al. 2012). Age plots for the *EDA* data were created by weighting all the data by age using a Gaussian kernel with a bandwidth of 20 years. To estimate the usage for the maps of ‘younger’ speakers, the mean of the Gaussian curve was set at 20 years and to estimate the usage of the ‘older’ speakers, the mean was set at 80 years. Kernel smoothing in space was then applied to these weighted data in the same way for other maps.
- 3 The wording of the question is subtly different in the Introduction to the Basic Materials (Orton 1962: 84) to that within the transcribed datasets of the *SED* and is part of a set of elicitations of different seasons and times of the day.
- 4 Interestingly, the *SED* informant from Nafferton, the Survey’s nearest site to Bridlington, used *autumn*.
- 5 Two out of the 11 users who self-identified as non-binary selected *backend*, a significantly higher proportion than in the sample as a whole. Given the small numbers, we do not wish to argue, however, that this difference would be robust if we had secured larger numbers of non-binary participants.

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